Demand Forecasting & Menu Optimization for Operational Efficiency in a Cloud Kitchen

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BDM Capstone Project, IITM Online BS Degree Program

Organisation Background

About the Business

Owner:	Mrs. Padmavathy Ravichandran
Platform:	Cookr App (B2C, delivery-only)
Location:	Chennai
Established:	February 2023
Cuisine:	30+ authentic home-style dishes
Meal Periods:	Breakfast, Lunch, Dinner
Order Types:	Regular (on-demand) + Monthly Subscription
Team:	Mrs. Padmavathy (cook) + helper + husband

Figure 1: Organisation Setup

Problem Statement

PS1

Unpredictable Demand → Emergency Procurement

- Daily order CV = 44.44% — demand cannot be predicted by intuition
- Stockouts force purchases on Quick Commerce
- Quick Commerce markup: Tomato +140%, Chicken +80%, Eggs +133%
- Margin erosion on every emergency purchase

PS2

Peak-Hour Surges → Unplanned Kitchen Closures

- Thursday = 17.64% of all weekly orders
- Lunch = 74.1% of all orders — 2-hour pressure window
- Cook exhaustion → forced closures → Rs. 9,718 revenue loss per closure
- No capacity management system in place

PS3

No Dish-Level Profitability Visibility

- 30+ dishes — no cost analysis ever done
- Ven Pongal sold at LOSS (–1.8% CM) — owner unaware
- Chicken Biryani (most popular) has only 29.2% CM
- No basis for pricing, promotion, or removal decisions

Data Collection

Process of Collection:

1. Understood business, explained project, obtained consent from Mrs. Padmavathy
2. Structured Q&A on ingredient costs, portion sizes, procurement habits, operational events
3. Manually transcribed Cookr app order history into Google Sheets day by day
4. Cost of Subscription Items were found from Cookr platform and by discussing with Mrs. Padmavathy

Four Primary Datasets:

1. **Order Details – 567 Orders, 49 operational days (April 1- May 26)**
2. **Menu Details – 114 Menu Items, Dishes description and selling price**
3. **Ingredient Costs – 86 Ingredients, Local vs Quick Commerce Cost**
4. **Operational Events - 15 Operational Events including Festival, holidays and closure periods**

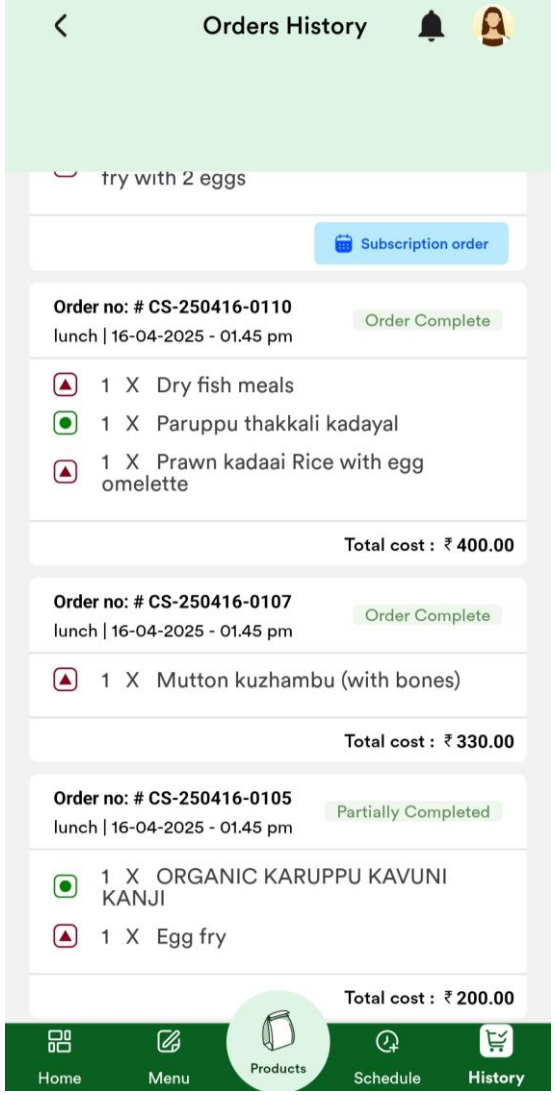

Figure 2: Cookr App Platform: Order History

Methodology

	PS1 — Demand Forecasting	PS2 — Demand Pattern Analysis	PS3 — Menu Profitability Analysis
Problem Statement			
Method & Tools	▸ 7-Day Moving Average ▸ Exponential Smoothing (α=0.3) ▸ MAPE = 25.51%	▸ Day-of-week grouping ▸ Meal period pivot ▸ Descriptive statistics	▸ Contribution Margin formula ▸ Menu Engineering Matrix ▸ 20 dishes analysed
Key Finding	• CV = 44.44% confirms high volatility • ES forecast tracks demand within 25%	• Thursday = 17.64% of weekly orders • Lunch = 74.1% of all orders	• Ven Pongal CM = –1.8% (LOSS) • Chicken Biryani CM = 29.2% (low)
Suggestions	**Sunday ES review → Mon/Tue bulk procurement schedule**	**Order caps + cross-train helper Batch prep by 10AM Thursdays**	**Remove Ven Pongal Reprice Biryani ₹190→₹210**

Results: Demad Patterns

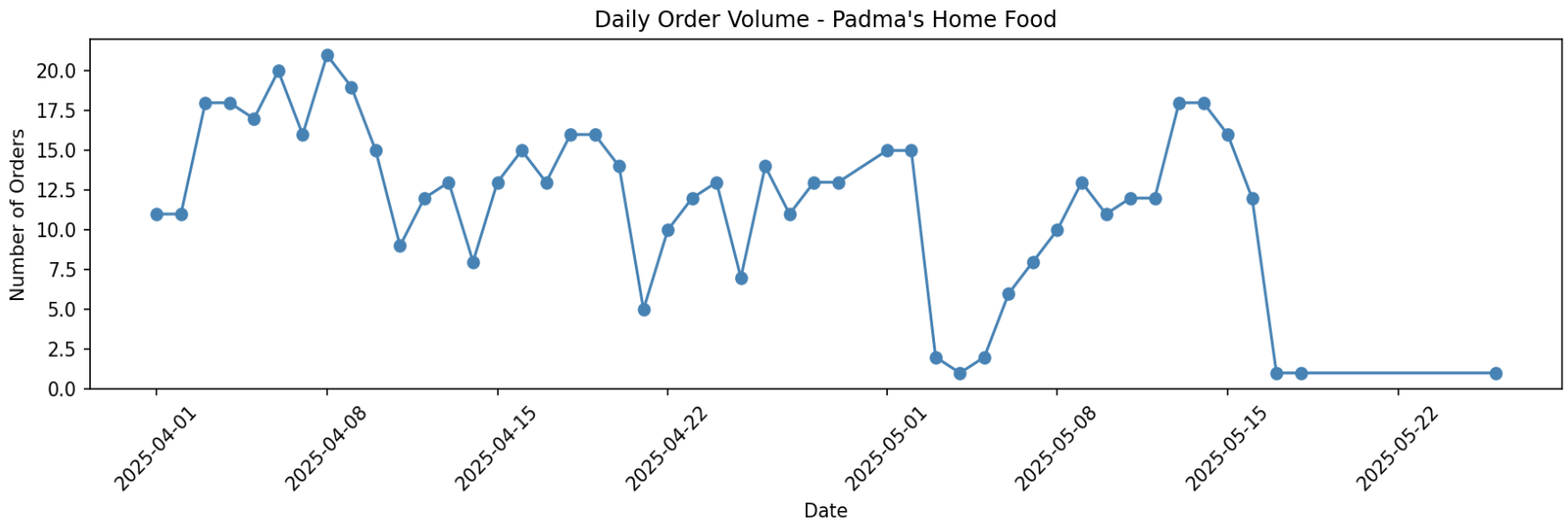

Figure 3: Daily Order Volume

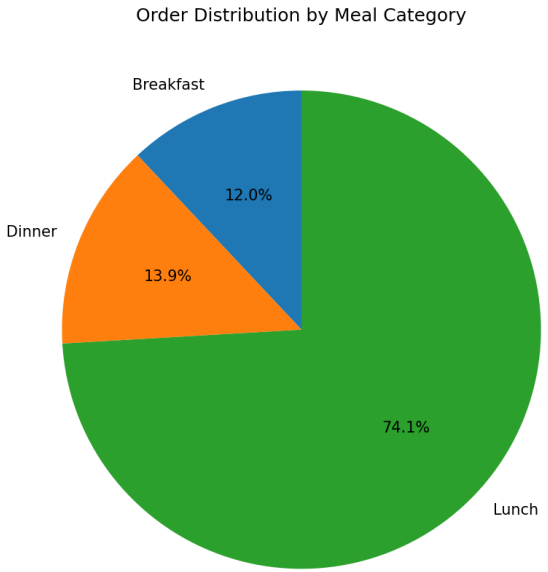

Figure 4: Meal Period Split

- **Thursday Peak: 100 orders = 17.64% of weekly volume**
- **Demand Volatility: CV = 44.44% — intuition-based procurement fails structurally**
- **Lunch Dominance: 74.1% of all orders in 2-hour window**

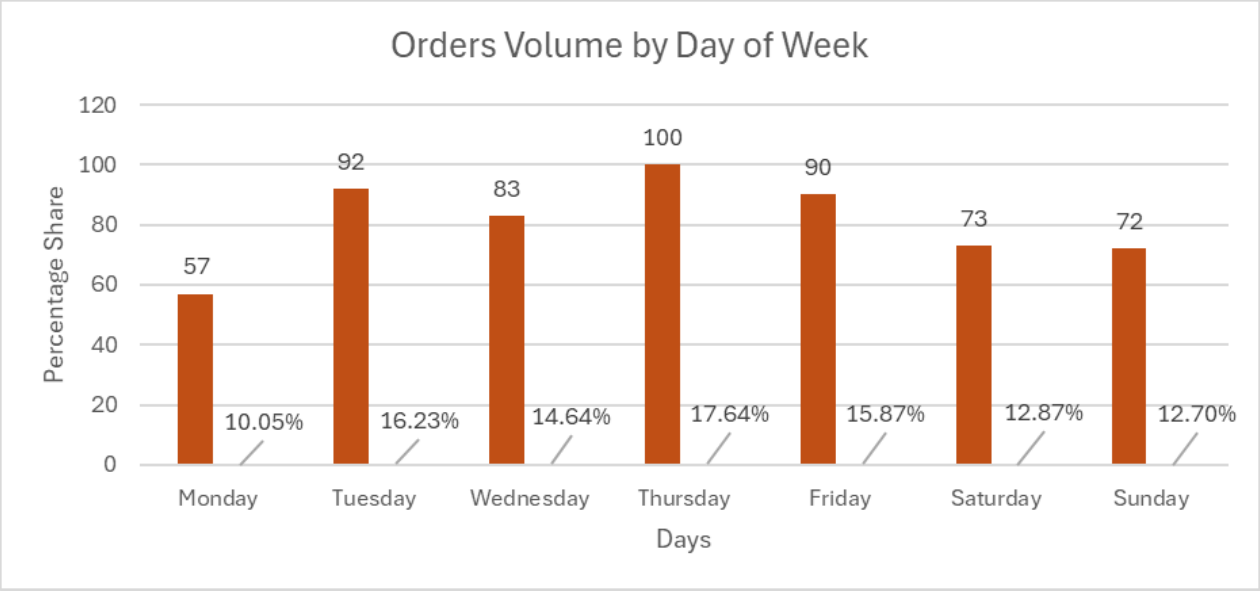

Figure 5a: Order Volume by Day of week

Results: Revenue Analysis

- 68.8% Revenue from Regular Orders
- The subscription menu and Rate cannot be changed
- Changing the regular menu's costing and removing unprofitable items and least ordered items like beverages

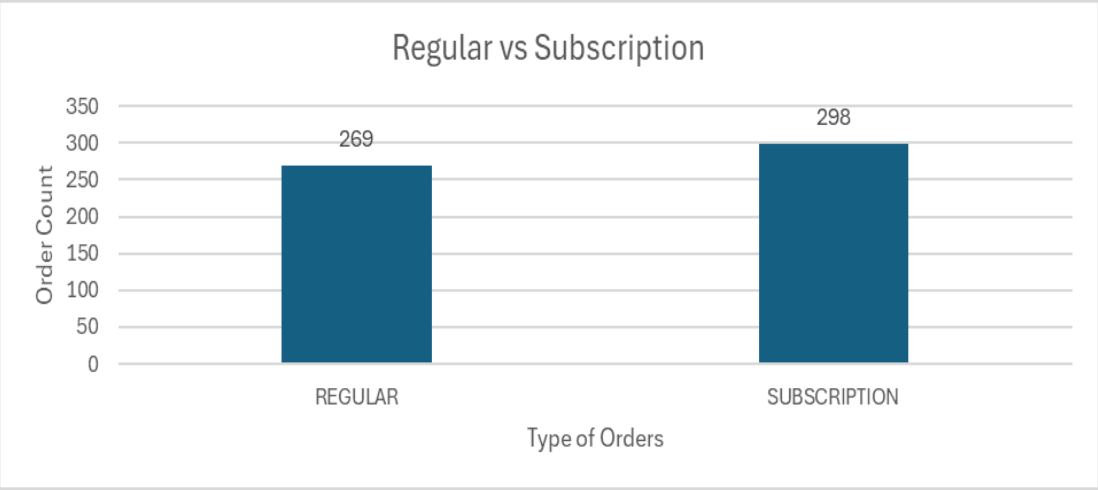

Figure 6: Orders Type (Regular vs Subscription)

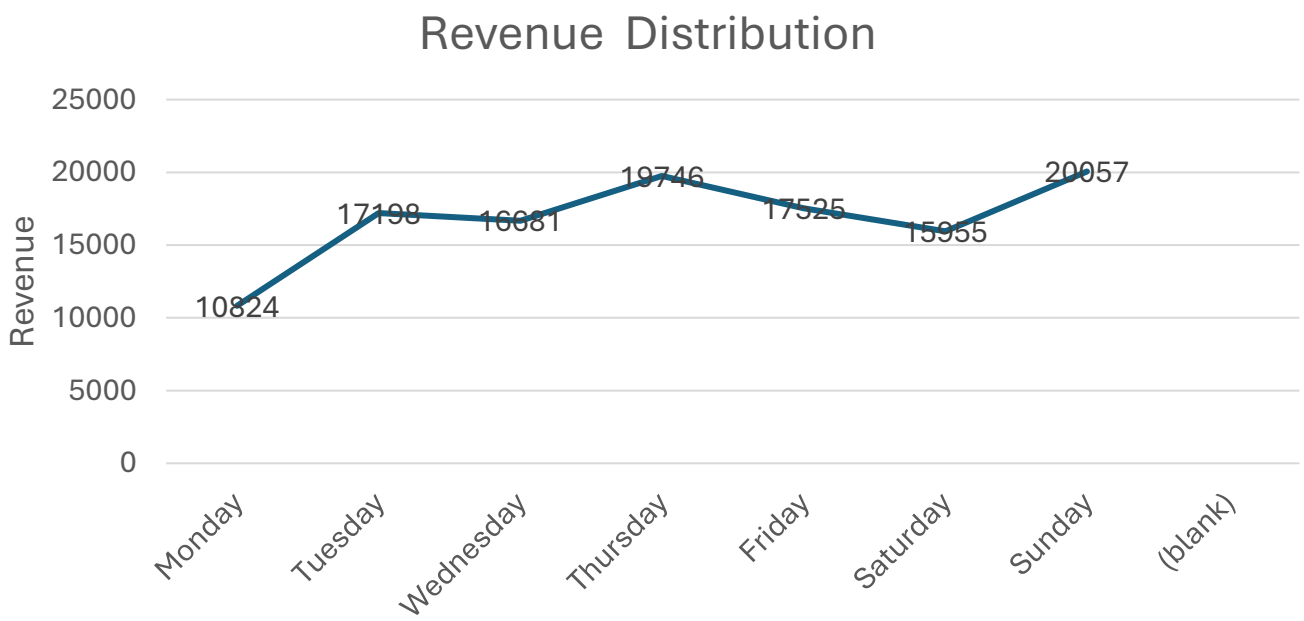

Figure 7a: Revenue Distribution across days

ORDER TYPE STATISTICS				
Row Labels	**NUMBER OF ORDERS**	**REVENUE**		
REGULAR	269	81216		
SUBSCRIPTION	298	36770		
Grand Total	**567**	**117986**		

% of Orders	Avg Order/ Day	% of Revenue
47.44268078	5.604166667	68.83528554
52.55731922	6.208333333	31.16471446
100	11.8125	100

Figure 7b: Revenue Distribution by order type

Results: Inventory Optimisation

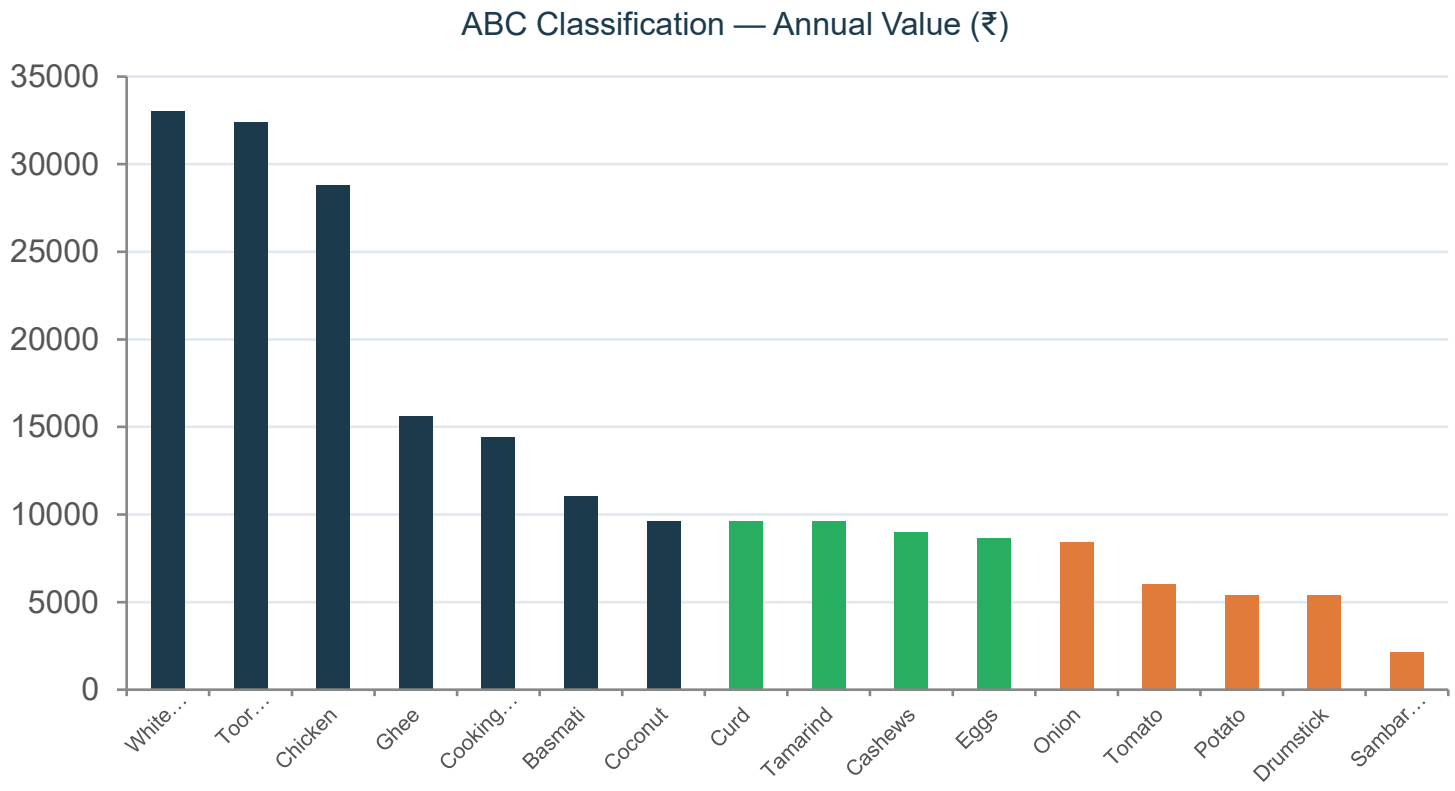

Figure 8: ABC Classification

Class A — 7 items

69.3% of annual ingredient spend

Class B — 4 items

16.2% of annual ingredient spend

Class C — 5 items

14.5% of annual ingredient spend

EOQ Example — Toor Dal

Current:	0.5 kg
EOQ:	31.84 kg/order
ROP:	1.47 kg in stock
Safety:	0.47 kg buffer

Quick Commerce Markup: Tomato +140% | Eggs +133% | Chicken +80% | Drumstick +80% | Potato +67%

Results: Menu Engineering Matrix

⭐ STAR

High Popularity + High CM

Keerai Paruppu Kadayal — 41.9% CM
Chicken Thokku — 45.8% CM
White Rice — 49.4% CM
Chicken 65 — 36.9% CM
Chapathi + Chicken Curry — 35.0% CM

❓ PUZZLE

Low Popularity + High CM

Vendakkai Poriyal — 48.0% CM
Rasam — 37.8% CM
Idly + Vada Curry — 37.0% CM
Lemon Rice + Egg — 34.0% CM
Pesarattu Dosa — 34.7% CM

🐴 PLOWHORSE

High Popularity + Low CM → Reprice

Chicken Biryani — 29.2% → Reprice ₹190→₹210
Vegetable Poriyal — 7.8%
Rice + Fish Gravy — 28.3%
Rice + Dhal + Chicken 65 — 25.7%

🐕 DOG

Low Popularity + Low CM → Remove/Review

Keerai Kootu — 21.4% CM
⚠ Ven Pongal — −1.8% (SOLD AT LOSS!)
Vegetable Poriyal — 7.8%

CM = Selling Price − Ingredient Cost − Platform Fee (30%) | 20 dishes analysed (10 regular + 10 subscription)

PS1 — Eliminate Emergency Procurement

- Sunday ES forecast → Mon/Tue bulk procurement
- ROP card for 7 Class A ingredients
- Local vendor only for Tomato + Chicken

Projected Savings ₹18-30k

PS2 — Control Peak-Hour Capacity

- Cross-train helper — rice + sambar (60% of lunch)
- Batch-prep by 10AM on Thursdays

Revenue fluctuations in closures avoided

PS3 — Optimise Menu Profitability

- Remove Ven Pongal (–1.8% CM = loss)
- Reprice Biryani ₹190→₹210 (CM 29%→38%)
- Promote Vendakkai Poriyal (48% CM) as add-on

Contribution to revenue